

Nathan Hammerschmitt Le Gal

nathan.p.hammerschmitt.le.gal.27@dartmouth.edu | (781) 584-2772 | www.linkedin.com/in/nathan-hammerschmitt-le-gal

Education

Dartmouth College, Hanover, NH

GPA: 3.83/4.0

Bachelor of Arts, Engineering Sciences modified with Studio Art

Jun 2027

Bachelor of Engineering, Mechanical Engineering

Jun 2028

Relevant Coursework: Solid Mechanics, Materials Science, Computer Aided Design, Embedded Systems, Digital Circuits,

Digital Fabrication, Physics, Differential Equations, Sculpture, Drawing

Programs: SolidWorks (CSWA certified), Fusion 360, Blender, Adobe Illustrator, Python, C, Matlab, Excel

Fabrication Skills: 3D printing, laser cutting, technical drawing, GD & T, tolerances, hand & power tools

Work Experience

Nexus Footwear Company, *Co-Founder & Mechanical Engineer*

Hanover, NH | **Jan 2025 – Present**

- Create a custom-fit, FDM-printed shoe built from digital scan data to improve comfort for users with foot abnormalities
- Design, produce, and test 50+ iterations, adjusting geometry, and print parameters to improve comfort and quality
- Assemble, modify, and maintain 6 FDM printers to build a print capacity of 80 pairs per month, automating G-code generation
- Survey 300 target customers and interview 50 target customers to identify failure modes and optimize print geometry
- Pitch and raise \$40k in R&D grant funding and venture capital, collaborating with co-founders on technical milestones

UMass Amherst Integrated Locomotion Lab, *Biomechanics Research Assistant*

Amherst, MA | **Jun 2024 – Aug 2024**

- Developed 30+ Python scripts to segment accelerometer data and extract angular-velocity-based running metrics
- Analyzed sensor outputs to link running shoe design and outdoor trail running biomechanics
- Assisted graduate students and postdoctoral researchers to facilitate motion-capture, IMU, and metabolic data collections

Dartmouth Admissions Office, *Writer*

Hanover, NH | **Sept 2023 – Present**

- Write weekly blog posts combining original writing and photography to represent the Dartmouth experience
- Drive 6k+ website visits to showcase engineering and entrepreneurial programs
- Curate compelling content to resonate with prospective students and coordinate weekly edits with admissions staff

Projects

Truss Bridge, Solid Mechanics

Aug 2025

- Led 3-person team to build plywood bridge (0.84 kg) that carried 20 kN (4.5k lbs) during Instron compression test
- Used Solidworks FEA and SAP2000 simulations to iterate on truss geometry and member size to maximize strength to weight
- Produced 2D drawings and fabrication plans, supervised cutting and assembly, and correlated test results with simulations

Dino Robot Walker, Computer Aided Design and Kinematics

May 2025

- Designed bipedal robot using a rotational 4-bar linkage with custom 3D printed and laser-cut parts
- Modeled linkage kinematics in GeoGebra to level foot path for efficient forward motion and stable stance
- Created assembly drawings and documentation for the mechanical system for repeatable fabrication and assembly

Leadership and Community Engagement

Magnuson Student Leadership Board, *Executive Member*

Hanover, NH | **Mar 2025 – Present**

- Facilitate guest speakers and panel discussions to share entrepreneurial expertise with 20+ students per event
- Led development of student startup fair to promote and connect 8 startups to 40+ students in attendance

Dartmouth Triathlon and Cycling Teams

Competitive Triathlete

Hanover, NH | **Sep 2023 – Present**

- Train 15 hours per week in-season to compete in local and national races
- Raced Triathlon Collegiate Club Nationals 2025, 7 sprint triathlons, 3 half marathons

Equipment Executive

Apr 2023 – Present

- Manage \$25k of team equipment for 40 active members to enable sport participation for all
- Designed and ordered Dartmouth-branded, Castelli performance kits, coordinating with suppliers on specs, sizing, and delivery

Additional Information

Interests: Additive Manufacturing, Automation, Robotics, Product Design, Entrepreneurship, Industrial Design

Other Experiences: VC Investment Competition (2024), Orientation Leader (2024, 2025), Fine Arts Student/Muralist (2014-2023)

Accomplishments: 4 Academic Citations (awarded to 2.4% of students), Leapyear Startup Accelerator (2025), Henry David Thoreau Scholarship (\$25k), Cell Signaling Technology Scholarship (\$10k), Harry Agganis Scholarship (\$8k), High School Valedictorian, first student accepted to Ivy League institution